5A Language of Algebra, Substitution and Like Terms

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Given $x = -3$, $y = 5$, and $z = -2$, evaluate each expression:

(a) $4y - 3x$ (b) $x^2 - 4z$ (c) $2x^2y - z^3$ (d) $\dfrac{y^2 - x^2}{z}$

Q2. Simplify each expression by collecting like terms:

(a) $8a - 3b - 11a + 7b$ (b) $6x^2 - 5x + 9 - 2x^2 - 4x - 12$ (c) $7mn - 4m^2n + 2nm - 5m^2n$ (d) $9pq^2 - 6p^2q - 12q^2p + 4qp^2$

Q3. Simplify each algebraic product:

(a) $-5a \times 6b$ (b) $-4x^2 \times (-7x^3)$ (c) $3m^2n \times (-8mn^2)$ (d) $(-2ab^2) \times (-3a^2b) \times (-4ab)$

Q4. Simplify each algebraic quotient:

(a) $\dfrac{-36x^4}{9x}$ (b) $\dfrac{-45a^3b^2}{-15ab}$ (c) $\dfrac{18m^2n^3}{-27mn^4}$ (d) $\dfrac{-6x^2y \times 4xy^3}{8x^3y^2}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A rectangular vegetable bed has length $(5x - 2)$ m and width $(2x + 3)$ m. (a) Write a simplified algebraic expression for its perimeter P. (b) Evaluate the perimeter when $x = 3$.

Q6. An AFL team kicks g goals (6 points each) and b behinds (1 point each) in the first half, and $(g + 3)$ goals and $(2b - 1)$ behinds in the second half. Write a simplified expression for the team's full-match score in points.

Q7. A rectangular sheet of metal has length $6a$ cm and width $4b$ cm. Four identical squares, each of side a cm, are cut from the corners. Write a simplified expression for the remaining area.

Q8. A plumber charges a call-out fee of \$85, plus \$70 per hour for h hours and \$15 per valve for v valves. Write an algebraic expression for the total charge and evaluate it when $h = 3$ and $v = 4$.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. Explain why $5x^2y$ and $-8yx^2$ are like terms, whereas $5x^2y$ and $5xy^2$ are unlike terms. Verify your explanation by substituting $x = 2$ and $y = 3$ into x^2y and xy^2.

Q10. Write an algebraic expression for the sum of four consecutive integers where the smallest integer is n. Simplify the expression and explain why the sum of four consecutive integers is never divisible by 4, but is always an even number.